04/12/24 M/S

USN | | | | | | | | | |

[23BSBE01]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

BIOLOGY FOR ENGINEERS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A	5x2=10		
1.	a.	List out the statements of cell theory.	L-1	CO1	02
	b.	Write any two examples of the kingdom of Protista.	L-1	CO2	02
	c.	Examples of organic and inorganic co-factors.	L-1	CO3	02
	d.	Discuss the laws of thermodynamics.	L-2	CO4	02
	e.	Write any four identification methods of microorganisms.	L-1	CO5	02
		PART - B	**6X5=30**		
2.	a.	Write the differences between prokaryotic cells and eukaryotic cells.	L-2	CO1	05
		OR			
	b.	What are proteins? List out the different types based on their structure.	L-2	CO1	05
3.	a.	How are heterotrophs classified? Briefly explain each.	L-2	CO2	05
		OR			
	b.	Briefly explain the classification of the kingdom Protista.	L-2	CO2	05
4.	a.	Describe the concept of activation energy with a neat diagram.	L-2	CO3	05
		OR			
	b.	Define the classification of cofactors with examples.	L-3	CO3	05
5.	a.	Why is ATP referred to as the energy currency?	L-2	CO4	05
		OR			
	b.	What are the steps involved in cellular respiration? Explain.	L-2	CO4	05
6.	a.	What is meant by microbiology? Briefly explain the importance of microbiology.	L-3	CO5	05
		OR			
	b.	What are the control measures for harmful microorganisms?	L-2	CO5	05
7.	a.	How are organisms classified based on the mode of ammonia excretion? Briefly explain each category.	L-2	CO2	05
		OR			

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

	b.	Explain exergonic reactions. Mention two examples.	L-2	CO4	**05**
		PART - C	**6X10=60**		
8.	**a.**	Write the various organelles present in prokaryotic and eukaryotic cells.	L-2	CO1	**10**
		OR			
	b.	Define chemotherapy and write a note on chemotherapy.	L-2	CO1	**10**
9.	**a.**	How are organisms classified into three domains based on molecular taxonomy? Briefly explain each domain.	L-3	CO2	**10**
		OR			
	b.	What are the general characteristics of members of kingdom of fungi and kingdom of plantae?	L-3	CO2	**10**
10.	**a.**	Summarize the structure of ribonuclease and lysozyme enzyme.	L-2	CO3	**10**
		OR			
	b.	Differentiate between lock/key model and induced fit hypothesis	L-2	CO3	**10**
11.	**a.**	Illustrate the process of cellular respiration.	L-3	CO4	**10**
		OR			
	b.	Summarize the second law of thermodynamics in biological Systems	L-2	CO4	**10**
12.	**a.**	Write a note on sterilization and growth methods of microorganisms.	L-2	CO5	**10**
		OR			
	b.	Mention the characteristics of protozoa and viruses. Provide examples for both.	L-2	CO5	**10**
13.	**a.**	Describe enzyme-catalysed reactions by taking digestion and DNA replication as an example.	L-3	CO3	**10**
		OR			
	b.	Explain the importance of the first and second laws of thermodynamics in biology.	L-3	CO4	**10**